Jude Crener Junior Pierre

Ph.D. Candidate in Information Science | Natural Language Processing (NLP), Medical NLP

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RESEARCH PROFILE

Ph.D. candidate at Nara Institute of Science and Technology (NAIST) specializing in medical Natural Language Processing (NLP), machine learning, and data-centric evaluation. My research focuses on building reliable NLP workflows for medical text, including word sense disambiguation, information extraction, named entity recognition, and LLM-based data augmentation in low-resource settings.

EDUCATION

Ph.D. in Information Engineering | Nara Institute of Science and Technology (NAIST), Japan | *Apr 2024 - Expected Mar 2027*

Focus: Medical NLP, information extraction, LLM-based data augmentation, and applied machine learning.

M.Sc. in Information Engineering | Nara Institute of Science and Technology (NAIST), Japan | *Apr 2022 - Mar 2024*

Focus: Medical NLP.

B.Sc. in Computer Science, Minor in Economics | Ecole Superieure d'Infotronique d'Haiti (ESIH) | *Sep 2014 - Jun 2018*

RESEARCH EXPERIENCE

Doctoral Researcher | Social Computing Laboratory, NAIST | *Apr 2024 - Present*

- Design and evaluate NLP pipelines for medical text, including preprocessing, annotation, word sense disambiguation, named entity recognition, and LLM-based data augmentation.
- Build experimental workflows for training and evaluating ML/NLP models in low-resource settings, with emphasis on data quality, semantic preservation, reproducibility, and reliable model evaluation.
- Communicate research through manuscripts, presentations, and collaboration with interdisciplinary research teams.

Research Assistant | NAIST | *May 2022 - Dec 2022*

- Performed data cleaning, annotation, and quality control for NLP datasets.
- Prepared structured datasets for machine learning experiments and analysis.
- Trained and evaluated NLP models and supported information extraction tasks.

SELECTED RESEARCH PROJECTS

- LLM-based data augmentation for rare disease named entity recognition: developed and evaluated synthetic medical text generation strategies designed to preserve entity labels and improve model performance in low-resource settings.
- Abbreviation expansion with minimal context: investigated word sense disambiguation for medical abbreviations using only a single word as context; accepted as a first-author paper at MEDINFO 2025.
- Medical information extraction workflows: experience with data preprocessing, annotation quality control, NER evaluation, and analysis of model errors for medical text.

PUBLICATIONS

International Conferences

- Pierre JCJ, Andrade GHB, Peng S, Yada S, Wakamiya S, Aramaki E. Abbreviation Expansion using only a Single Word as Context. In: Proceedings of the 20th World Congress on Medical and Health Informatics (MEDINFO 2025), Taipei, Taiwan, 2025. Read article: https://ebooks.iospress.nl/doi/10.3233/SHTI251228

Journal Papers

- Pierre JCJ, Nishiyama T, Peng S, Wakamiya S, Aramaki E. Persona-Driven Data Augmentation for Disease Name Recognition. JMIR Medical Informatics (5.0 IF). Manuscript Accepted (to be published soon). Read preprint: https://preprints.jmir.org/preprint/87831/accepted

HONORS AND AWARDS

- MEXT Japanese Government Scholarship Awardee, 2021 - present. Selected for the Japanese Government Scholarship for academic excellence.

ACADEMIC EXPERIENCE

- Teaching Assistant, NAIST: supported graduate-level NLP coursework, including classroom preparation, coding tasks, and Q&A support for students.

INDUSTRY EXPERIENCE

Communication Protocol Design Engineer, R&D Internship | Schneider Electric, Osaka | *Jan 2023 - Mar 2023*

- Designed and implemented an MQTT-based communication protocol for industrial HMI systems using C/C++.
- Developed communication functions for sending and receiving data between HMI devices and cloud-based systems.
- Worked on reliable data exchange with attention to performance and system-level constraints.

Business Risk Analyst | Digicel Group | *2018 - 2021*

- Conducted data analysis to identify and mitigate financial and operational risks.
- Produced reconciliation reports and analytical summaries to support business decision-making.

Web Developer, Freelance | Remote | *2017 - 2018*

- Developed web applications with frontend and backend components.
- Built backend services using JavaScript, Node.js, Python, and Flask-based frameworks.

TECHNICAL SKILLS

Programming	Python, JavaScript, C/C++, SQL
ML / NLP	PyTorch, Keras, Scikit-learn, transformer-based NLP models, model training and evaluation
Data Processing	NumPy, Pandas, data cleaning, annotation, quality control, experimental analysis
Visualization	Matplotlib, Seaborn
Web / Systems	Node.js, Flask, HTML, CSS, MQTT-based communication protocols

LANGUAGES

French: Native | English: Professional | Japanese: Basic